VitalScan – RF Human Detection System Final Report

ECE 510 – Instructor: Dr. Jafar Saniie

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About Us

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B.S./M.S. Telecommunications Engineering
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Experience in System-Level and Embedded Software

Problem Statement

The Challenge of Zero-Visibility Rescue

- First responders frequently operate in highly volatile, zero-visibility environments during structural fires and natural disasters.
- The ability to detect human life behind physical barriers is the definitive factor between rescue and recovery.
- While "sense-through-wall" technology exists, severe economic barriers keep these tools out of the hands of most emergency personnel.
- Rescue teams are often forced to rely on slow, hazardous, and ineffective manual search methods

Problem Statement

Firefighter Hazards and Visibility Limitations

- Firefighters must regularly enter high-temperature, smoke-filled rooms where line-of-sight is completely eliminated, due to smoke inhalation people lose their ability to ask for help while still conscious.
- Thermal imaging cameras often fail because ambient heat masks the thermal signature of a human body.
- Approximately 100,000 to 180,000+ people die annually from direct fire-related incidents worldwide, with 300,000–350,000 suffering injuries.
- In 2024, 72 firefighters lost their lives while on duty in the United States.

Our Solution: SDR-Based Human Detection

- We propose a portable RF-based human detection and occupancy estimation system.
- The system utilizes the ADALM-Pluto Software Defined Radio (SDR) coupled with open-source processing frameworks.
- This approach disrupts the current cost paradigm by shifting complex RF operations from rigid hardware to flexible software.

Market Analysis
Existing non-LOS human detection devices

Device	Primary Technology	Target Market	Estimated Unit Cost
Camero Xaver 400	UWB Pulse Radar	Elite Military / SWAT	~$47,500
Seek FirePRO 300	Thermal Imaging	Fire Rescue	$1,200
NASA FINDER	Microwave Radar	Federal Disaster Response	High/Proprietary Licensing
Our Proposed System	SDR (ADALM-Pluto) FMCW	Local/Volunteer Responders	< $300 (Total Prototype)

RF Doppler Principle for Human Motion Detection

- When a 5.8 GHz RF signal reflects off a moving human, the frequency of the returned signal shifts slightly. This shift allows us to detect motion even in zero-visibility environment
- Where: v = human velocity and λ= wavelength. Faster motion → larger Doppler shift. No motion → near-zero Doppler shift.
- Walking/Running victim → strong Doppler peak.
- Crawling victim → moderate Doppler energy.
- No movement → no Doppler energy.

Case-I : Moving Reflector/Target

Doppler Frequency = $\frac{2*v}{\lambda}$

Doppler Frequency = $\frac{2 * V_r}{\lambda} = \frac{2 * V_r * F_{tx}}{C_0}$

V = Speed of wave source (m/sec)

λ = wavelength (m)

Vr = radial speed of aim

C_0 = velocity of light in free space

Sample Doppler Calculation

- Assumptions: Victim motion scenarios inside a burning structure:
  - Crawling victim speed = 0.5m/s
  - Walking/Stumbling victim = 1.2m/s
  - Radar transmit frequency = 5.8875GHz

Calculation: $\Delta f = 2fv/c$

Δf : change in frequency

f: transmitted frequency

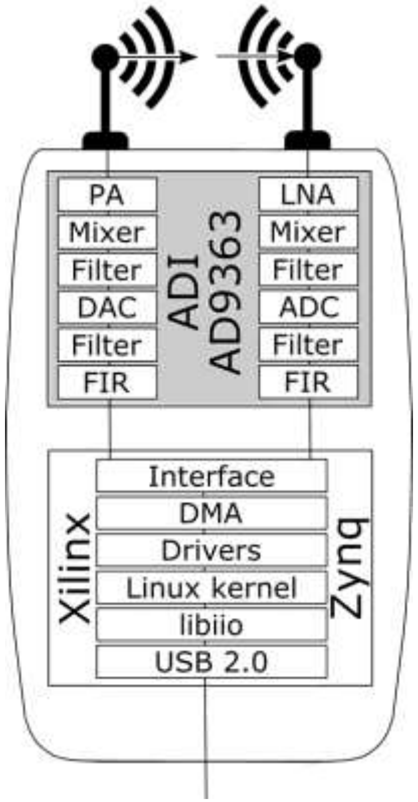
V: velocity of observed object

C: speed of EM waves

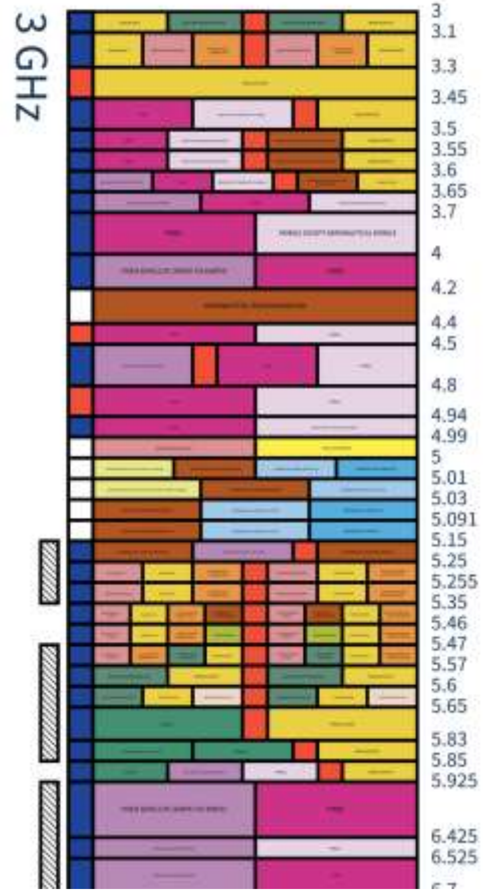
Slowdown	Speed	70 MHz	5.8875 GHz	6.0 GHz
0%	0.5m/s	0.233 Hz	19.625Hz	20Hz
0%	1.2m/s	0.560Hz	47.1Hz	48Hz
50%	0.5m/s	0.467Hz	39.250Hz	40Hz
50%	1.2m/s	1.120Hz	94.200Hz	96Hz
80%	0.5m/s	1.167Hz	98.125Hz	100Hz
80%	1.2m/s	2.8Hz	235.5Hz	240Hz

Hardware – ADALM Pluto

- Acts as the bridge between our physical RF environment and the digital signal processing.
- Uses SDR while being cost-effective and portable.
- Specifications:
  - Chipset: modified via Firmware to AD9361
  - Freq. Range: **70 MHz - 6 GHz**
  - Max BW: **56 MHz**
- It has been Hardware modified to unlock the second Rx port via a U.Fl to SMA internal cable modification.



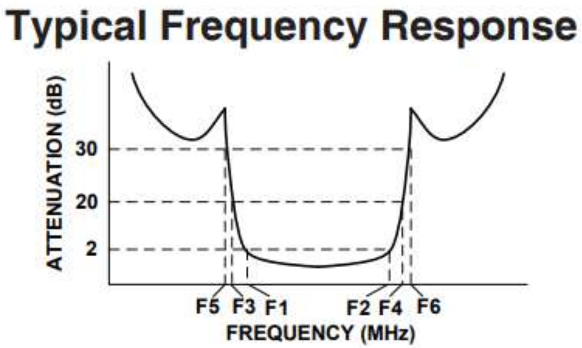
Hardware – RF System Design & Frequency Selection



(US frequency allocation)

- **Bistatic Radar** design with **1Tx-2Rx**.
- **Operational central frequency** will be **5.8875 GHz** to avoid interferences.
- Used **BPFs** on the Tx and 2Rx to isolate the signal from the interferences.
  - High rejection (30dB up to 17GHz)
  - Low Insertion Loss (1.26dB)
- Tx **Amplifier** (gain of **28 dB** in the 5.8 GHz ISM band
- **LNAs** on the RXs to improve SNR (23 dB gain at 5.8 GHz and NF of 2dB).

VBFZ-5500-S+BPF (4.9 – 6.2 GHz)



EVAL-CN0523-EBZ

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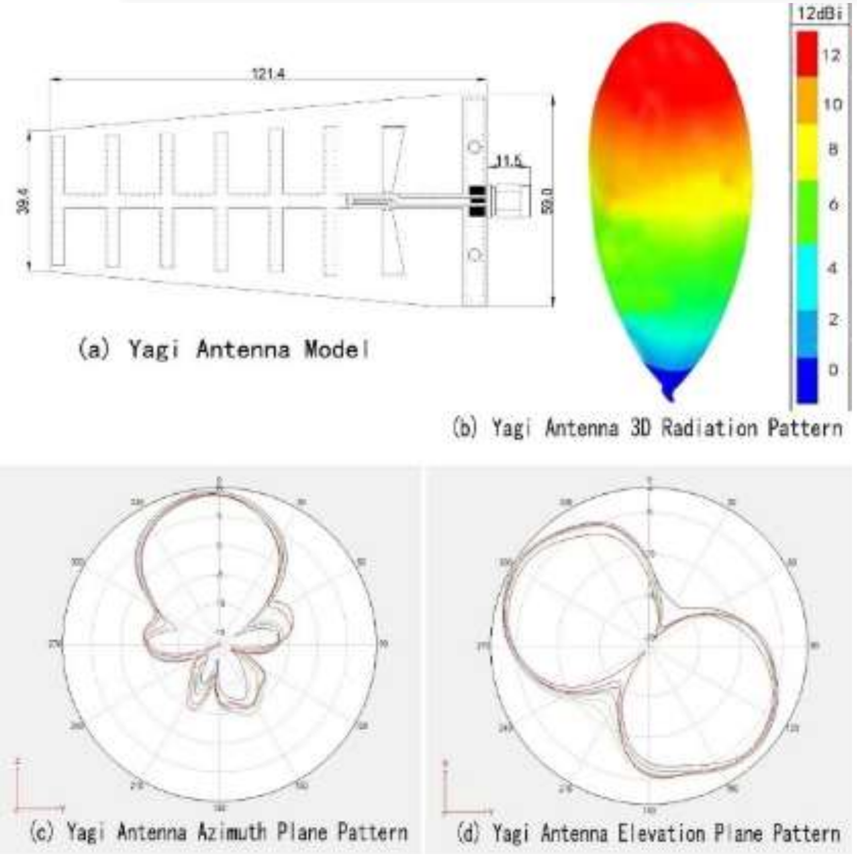
Hardware – Antennas (Planar Yagi Antenna)



(a) Yagi Antenna Model

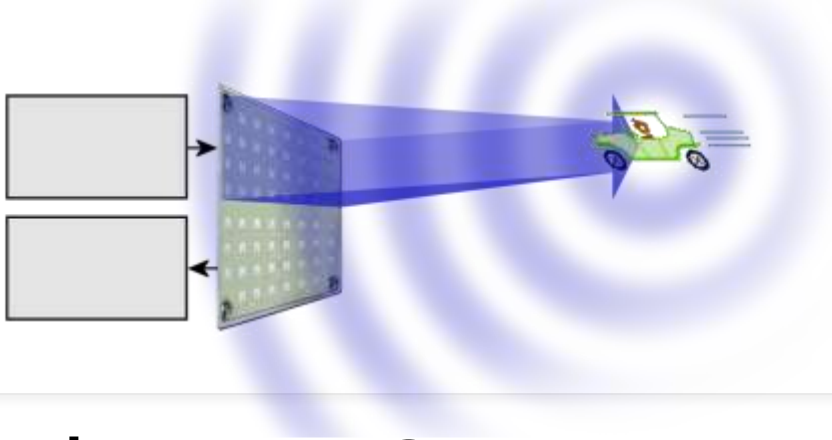
(b) Yagi Antenna 3D Radiation Pattern

(c) Yagi Antenna Azimuth Plane Pattern

(d) Yagi Antenna Elevation Plane Pattern

- **Key Specifications:**
  - **Tuned Frequency Bands:** 2.4 GHz and 5.8 GHz ISM bands
  - **Peak Gain:** 12 dBi
  - **Impedance & Connector:** 50 Ω with RP-SMA connector (adapter used for ADALM-Pluto)
  - **Radiation Pattern:** Highly directional with a narrow forward lobe
- **Benefits for our System:**
  - Dramatically improves wall penetration and maximum detection range.
  - Actively rejects unwanted lateral room clutter.
  - Provides a much cleaner baseband signal for DSP algorithms to extract human movement.

CW Doppler



Continuous Wave (CW) Doppler Radar

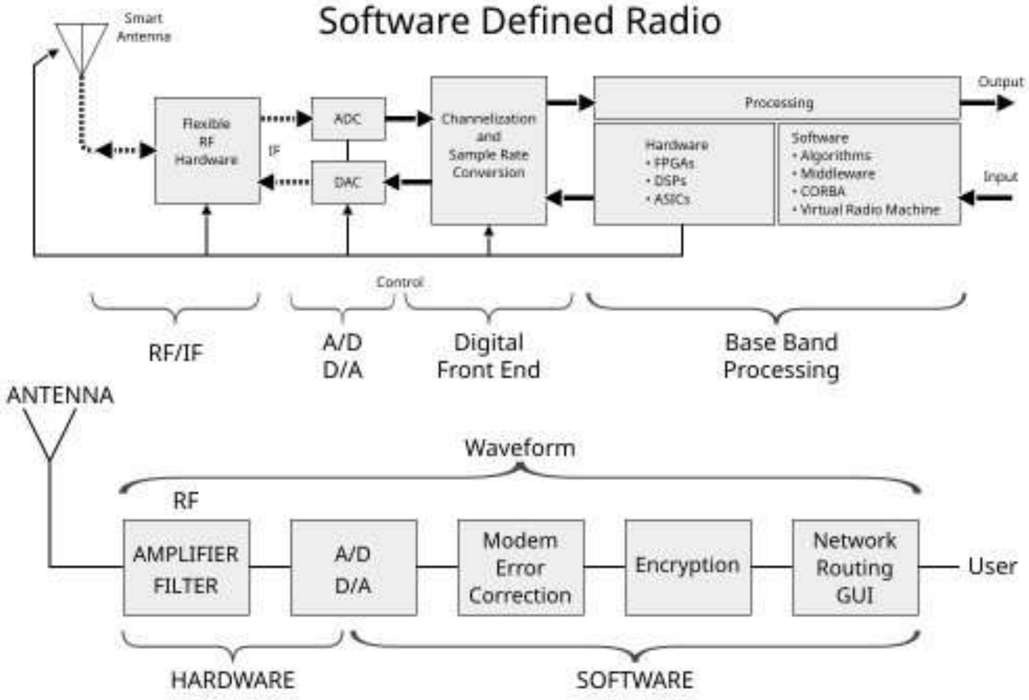
The current VitalScan prototype uses CW Doppler radar to detect human motion.

- **Key Concept:**
    - o The radar transmits a continuous signal and measures Doppler frequency shift caused by moving objects.

Advantages for Fire Detection

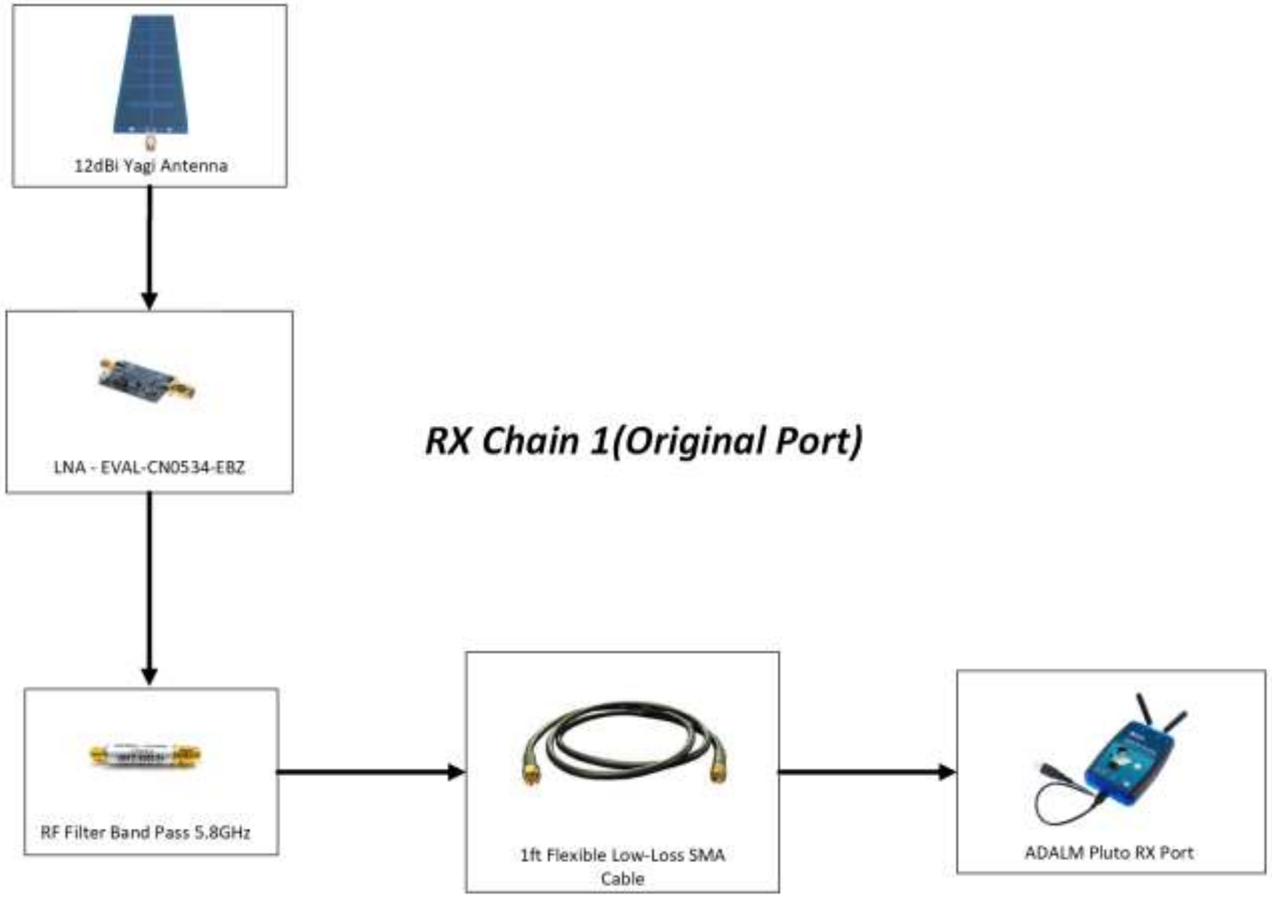
- Simple hardware implementation
- Low computational complexity
- Reliable motion detection through smoke or walls
- Detects crawling, walking, and subtle human movement

SDR



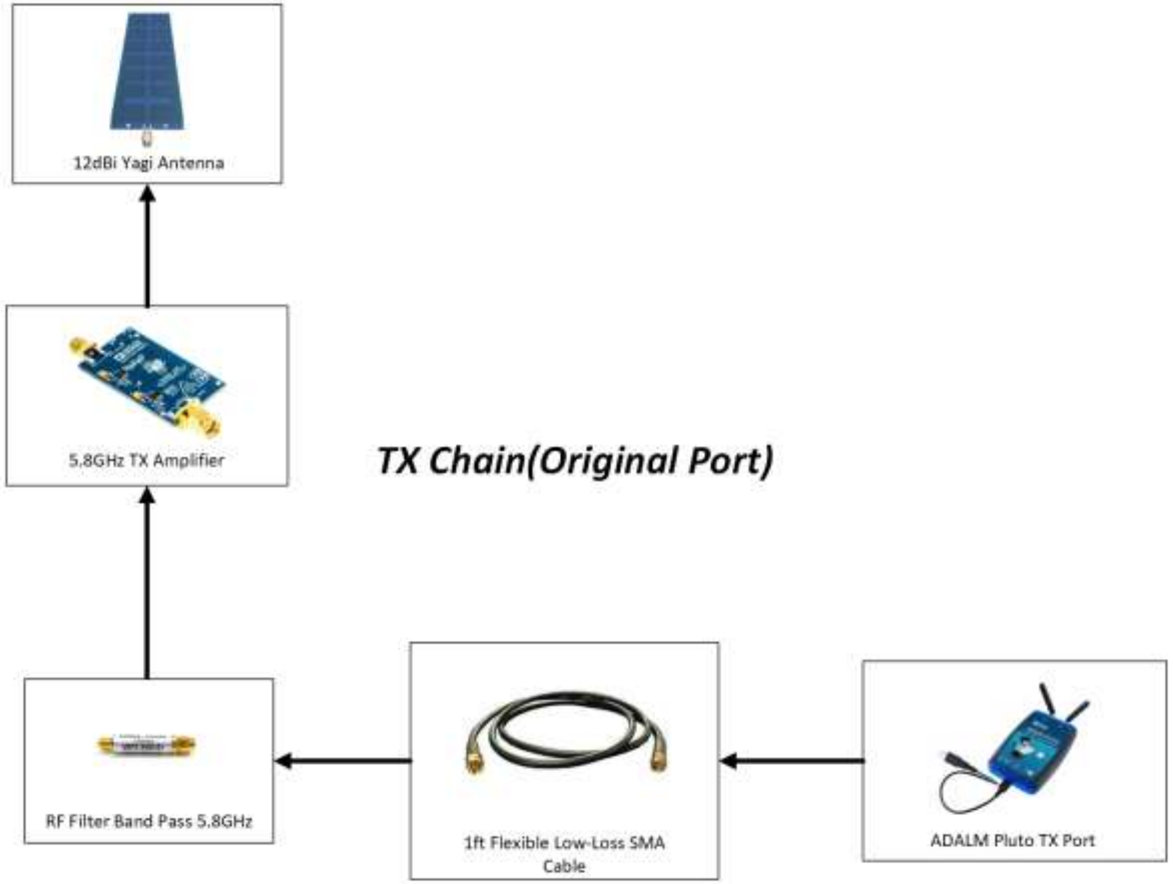
Signal Processing Pipeline(RX Chain)

Goal: convert raw IQ into **motion state** + **occupancy/density estimate** (coarse zone heatmap as well if we have enough time)



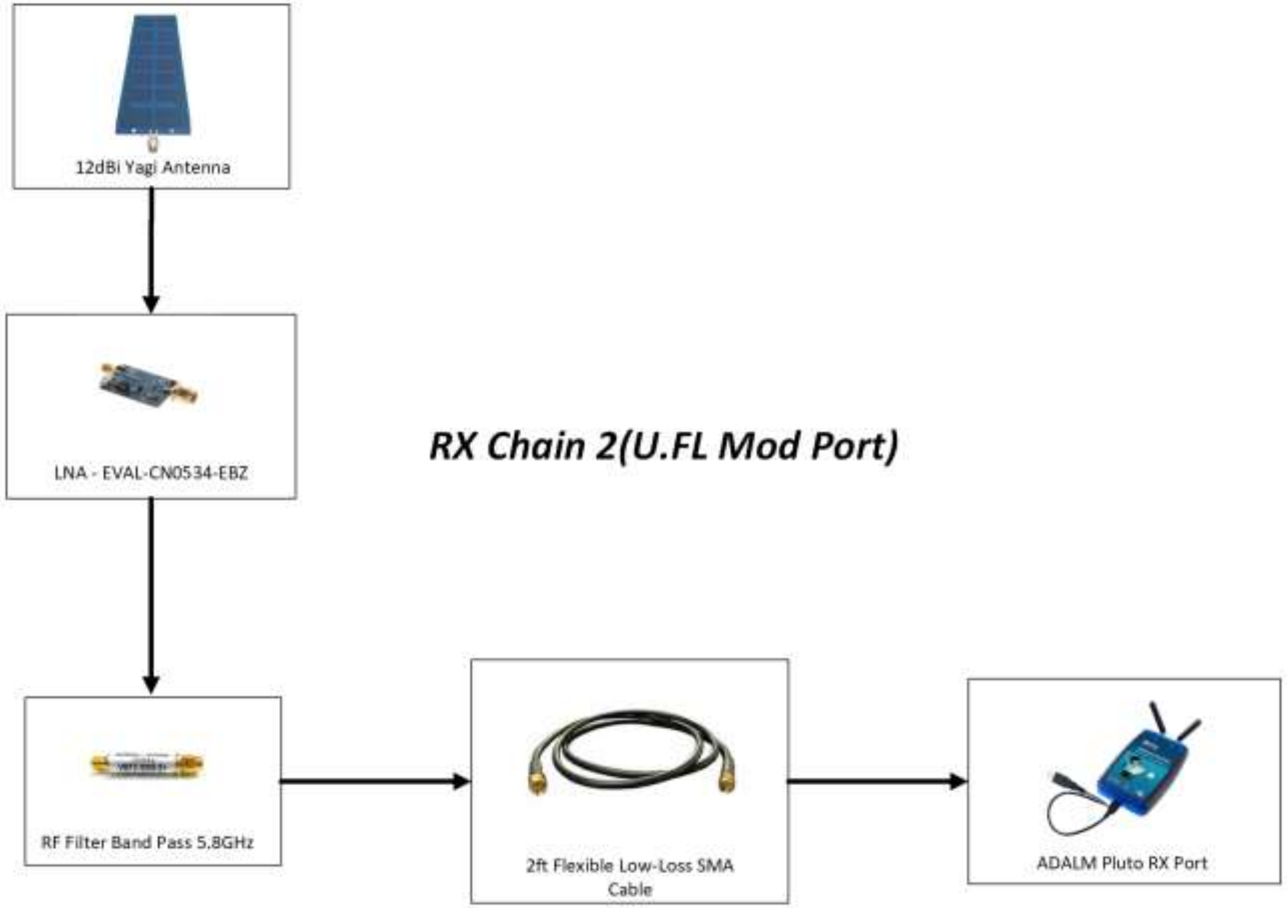
Signal Processing Pipeline(TX Chain)

Goal: convert raw IQ into **motion state** + **occupancy/density estimate** (coarse zone heatmap as well if we have enough time)



Signal Processing Pipeline(2nd RX Chain)

Goal: convert raw IQ into **motion state** + **occupancy/density estimate** (coarse zone heatmap as well if we have enough time)



Implementation Stack & Scope Evolution

Software & Tools

Language: Python 3

SDR Library: pyadi-iio + libiio

DSP: NumPy / SciPy

Display: Matplotlib

Hardware: ADALM-Pluto (AD9361)

Protocol: USB 3.0 → IIO backend

Future: Raspberry Pi 5 edge deploy

Hardware Constraint & Scope Change

Original goal: detect breathing (0.3 Hz chest motion)

- Required Doppler resolution of ~0.4 Hz at 5.8875 GHz
- SNR at -20 dBm ERP was insufficient to resolve
- Breathing above the TX leakage noise floor

Revised scope: detect motion (walking, crawling, waving)

- Doppler shifts of 10–40 Hz are clearly detectable
- System validated at up to 6 ft with LNA + TX PA

Through-Wall Detection – Demo Video

Analysis Of Demo Video – Output explanation

1st Scenario

2nd Scenario

3rd Scenario

System Architecture & Design

Goal: Consolidate the "Compute-First" and "Local-First" strategies while grounding the software in the physical design.

- **Software Pipeline:** Transitions from high-bandwidth **GNU Radio** acquisition on a laptop to optimized **NumPy/Liquid-DSP** on the **Raspberry Pi 5** for edge processing.
- **Detection Strategy:** Uses a **Moving Target Indicator (MTI)** filter to focus on 0.5Hz–10Hz Doppler shifts, effectively isolating human movement from static clutter.
- **Wireless Transmission:** The Raspberry Pi 5 hosts a self-contained Wi-Fi hotspot and streams raw IQ data to a connected laptop, running the full signal processing chain without an existing network infrastructure.

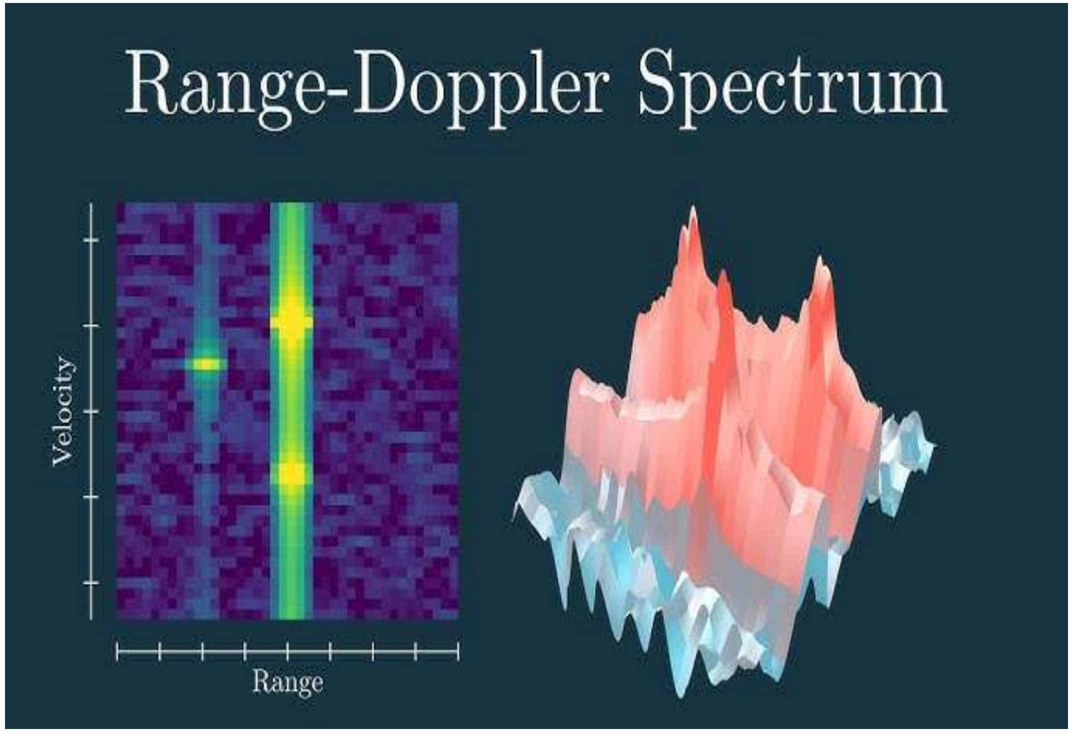
2 Stage Detection Algorithm

Stage 1: Motion Classifer:

Computes Weighted Z-Score features against a CFAR empty baseline. Phase RMS (1-60 Hz) acts as the primary discriminant, mapping a composite score to confidence via Sigmoid logic.

Stage 2: Clutter Rejection

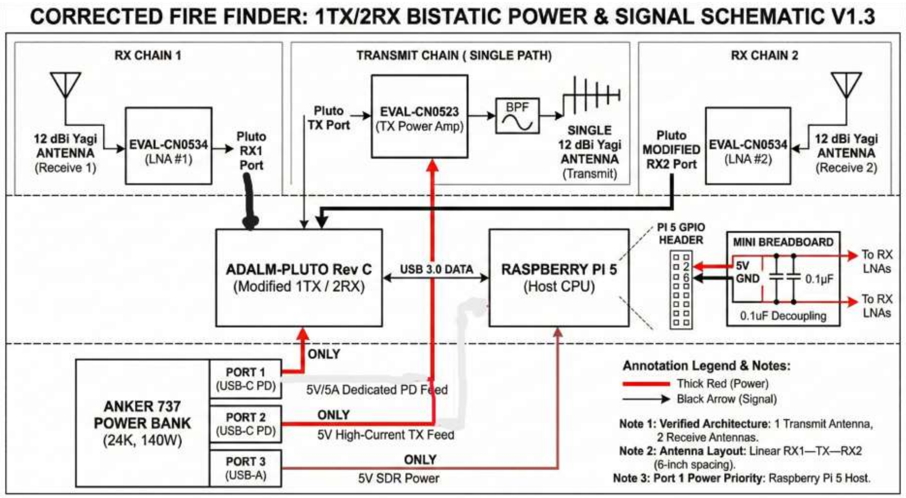
Applies soft penalties for mechanical sources: High-Phase-Noise (>10 Hz gate), Spectral-Width (<30 Hz RMS), and Peak-Persistence (>70% bin stability) to reject fans and HVAC units.



Power Analysis & Distribution

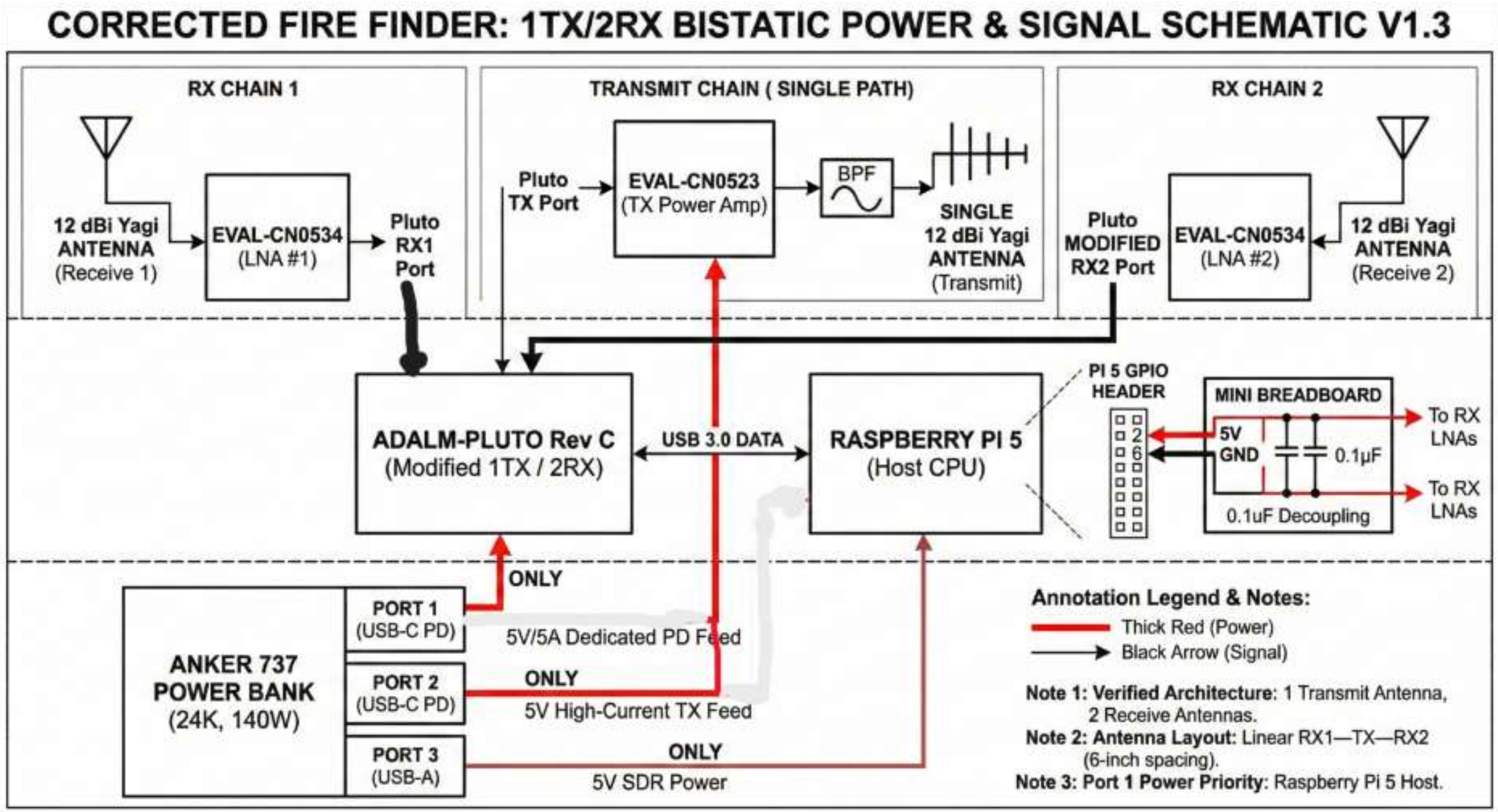
Goal: Address the high power demands of the Pi 5 and RF amplifiers to ensure "tactical readiness."

- **Total Power Requirement:** The system draws **21W–33W** (~4.3A–6.7A @ 5V), necessitating a high-wattage source like the **Anker 737 Power Bank**.
- **The 5A Constraint:** The Pi 5 requires a dedicated **5V/5A** supply; without the proper USB-C PD handshake, it throttles performance, which would jeopardize the <1s response time requirement.
- **Noise Mitigation: Decoupling capacitors** (100uF + 0.1uF) are integrated across TX/RX amplifier inputs to smooth switching regulator noise that could otherwise manifest as "false motion" in the FFT.

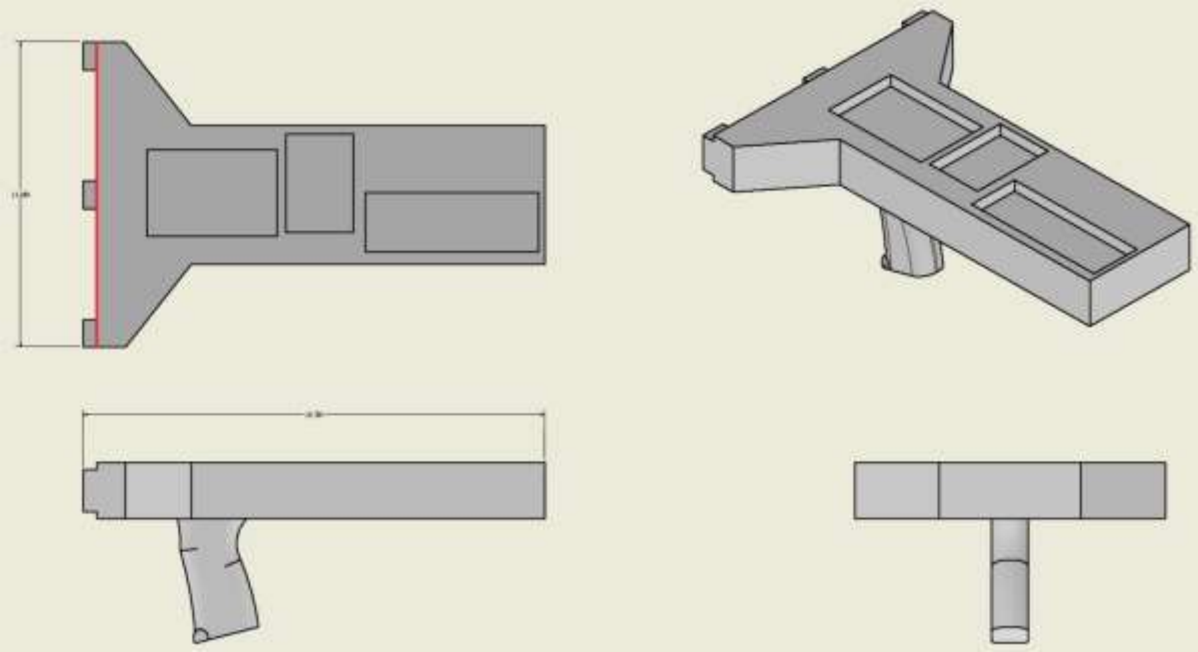


Enlarged Version on Next Slide

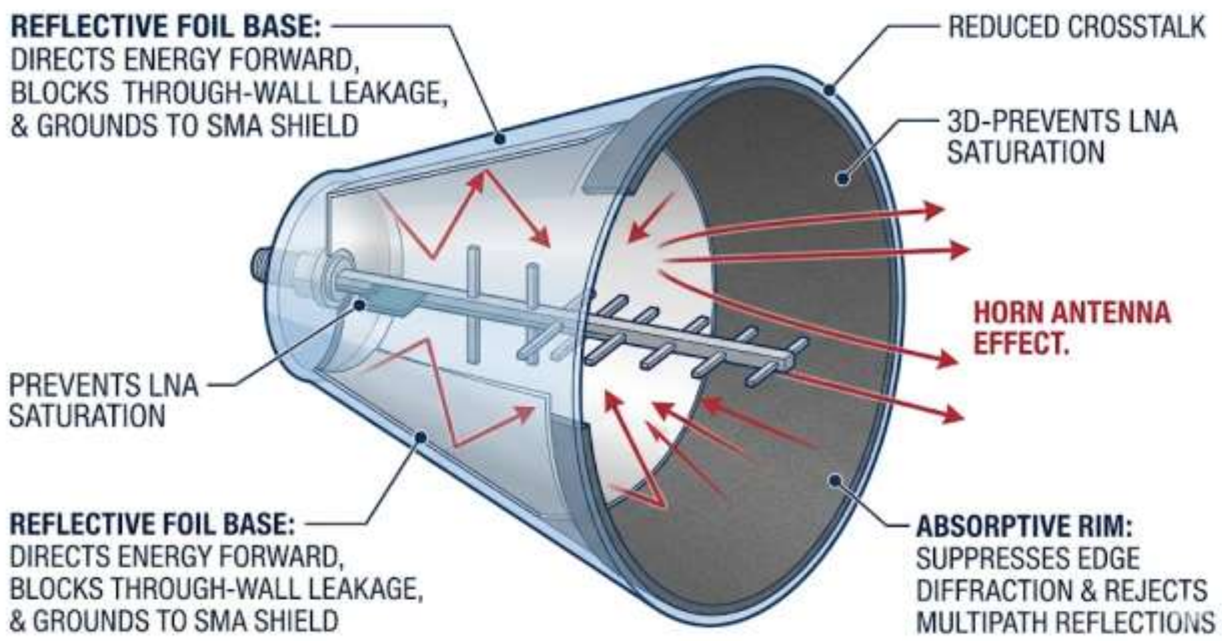
Power Analysis & Distribution Block Diagram



Product Design: Prototypes



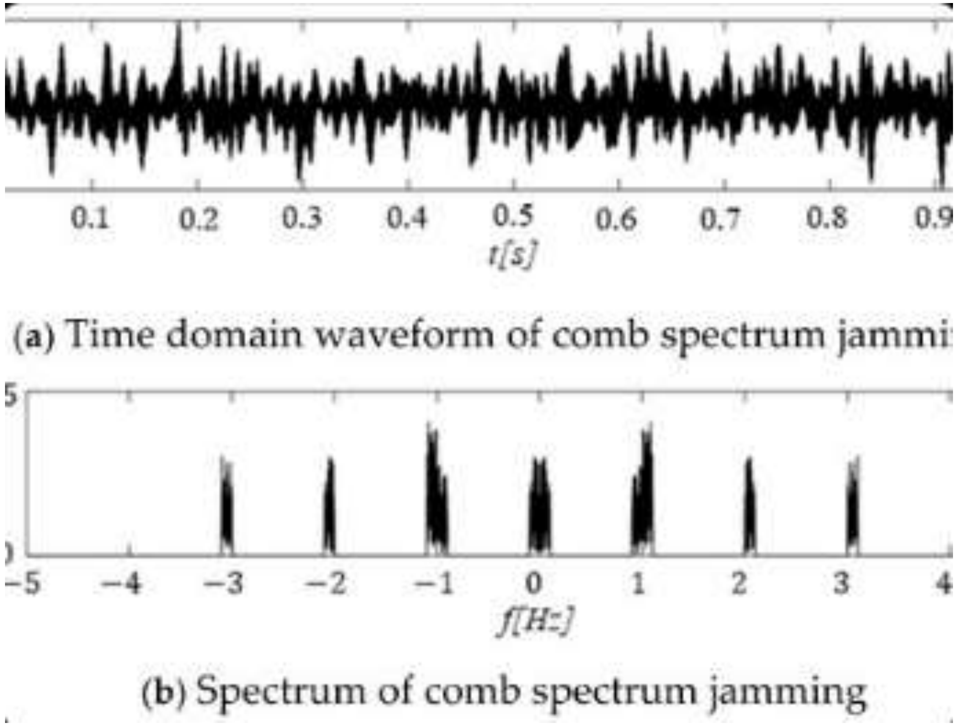
Product Design: Cones



Security & Signal Constraints

Signal Vulnerabilities

- RF Jamming: Targeted high-power interference on the 5.8875 GHz carrier can overwhelm the LNA, causing complete detection failure.
- Bandpass Traffic: High traffic in common ISM bands can saturate our bandpass filters, masking human Doppler signatures within the noise floor.
- Electronic Denial: Malicious saturation of our bandwidth results in "Blind Zones," critical for tactical rescue missions.



(a) Time domain waveform of comb spectrum jammi

(b) Spectrum of comb spectrum jamming

Team Roles/Contribution

Adrian Korwel

Team Captain & Primary Tester. Led project theory research, developed DSP filters, and managed overall hardware/software integration.

Elijah Johnson

Computing Lead. Specialized in Raspberry Pi 5 optimization, research, and high-performance computation strategy for edge processing.

Bryan Evans

Hardware & Design Lead. Managed power distribution, component selection, housing design (CAD/3D printing), and system testing.

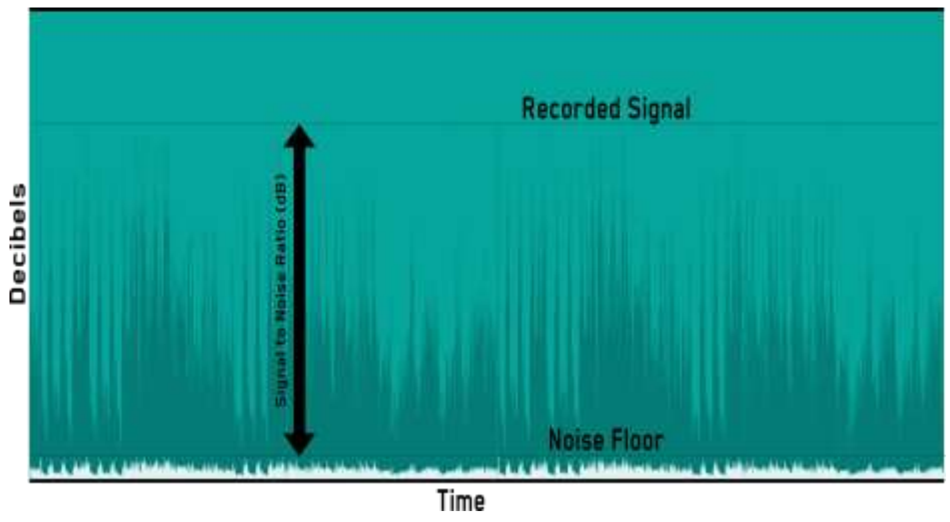
Alvaro Tenorio

Research Specialist. Focused on project theory research, RF physics, and signal propagation analysis for through-wall scenarios.

Key Project Challenges

Key Project Challenges

- **Hardware Constraints:** Found at the beginning of Phase 2 that the Pluto's internal noise and 5.8 GHz limit prevent reading micro-Doppler movements like breathing, necessitating a shift in our primary success metrics.
- **Isolation vs. Portability:** To reduce RF crosstalk, the antenna "shrouds" (cones) were enlarged, doubling the expected footprint of the handheld chassis and complicating the "tactical" portability goal. This resulted in greater RF shield usage.



Project Timeline (Phased Approach)

Phase 1: Foundation & Validation (Completed)

- Procured the ADALM-Pluto and 12 dBi Yagi antenna array.
- Successfully implemented the CW Doppler baseline; validated mo target detection against human motion in open-air environments.

Phase 2: Hardware "Hacking" & Scope Pivot (Completed)

- Modified Pluto firmware to enable the 2nd Receive channel (2 RX, 1 TX) to support future interferometry testing.
- **Scope Adjustment:** Research and initial bench testing revealed the **ADALM-Pluto noise floor** is too high to reliably resolve **micro-Doppler** signatures, such as human breathing; the project scope was narrowed to prioritize robust detection of active motion (crawling/walking) behind barriers.

Phase 3: Barrier Testing & Portable Optimization (Completed)

- **Software Port:** Translated the original GNU Radio signal processing filters into an optimized Python-based pipeline (using NumPy/Liquid-DSP) for the portable version.
- **Design Hurdle:** Portable chassis design in Autodesk Inventor became a major challenge as the overall unit size grew significantly to accommodate the **flared cone antennas** required to eliminate the high TX-to-RX leakage observed in initial testing.

Phase 4: Portable Integration & Stress Testing (Current)

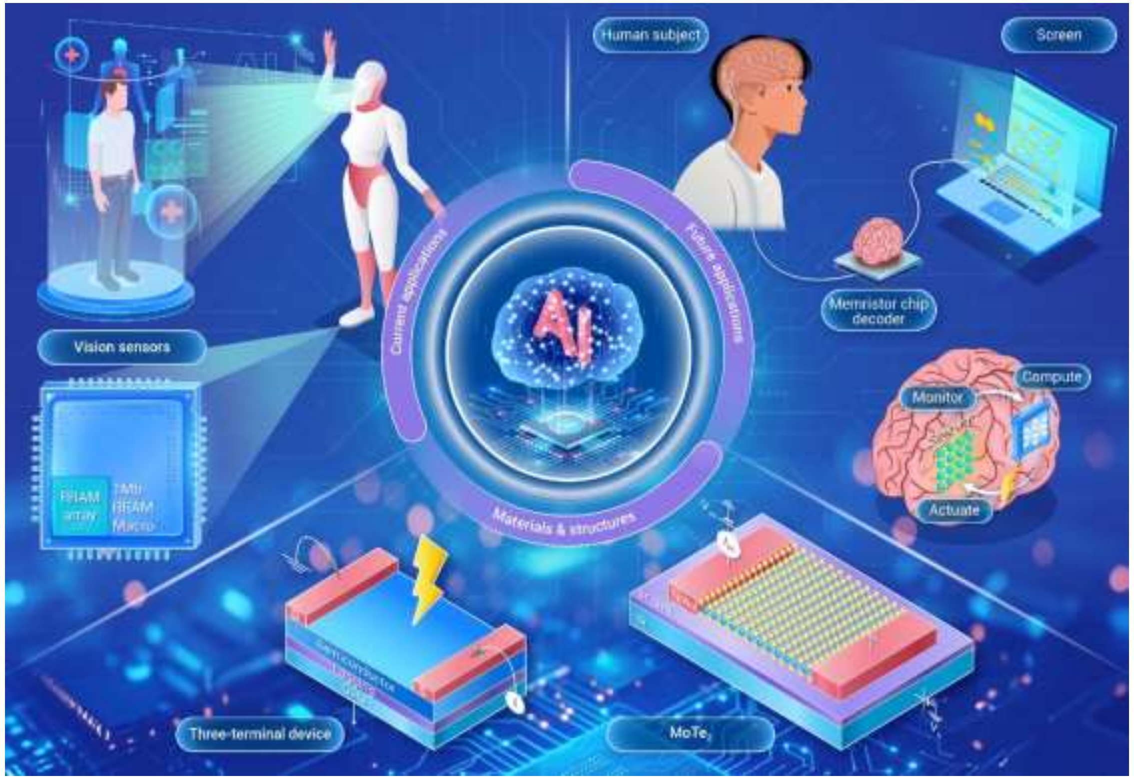
- Finalizing the transition of the Python-based MTI pipeline onto the Raspberry Pi 5.
- Conducting stress tests to ensure the Pi can maintain <1s tactical refresh rates without thermal throttling or lag under peak current draw.

Future Works

Tactical Portability
Redesigning the chassis with light-weight, RF-absorbent materials to reduce the "horn" footprint while maintaining RX/TX isolation.

Hardware Accuracy
Transitioning from ADALM-Pluto to high-SNR dedicated FMCW transceivers to re-enable sub-Hz breathing and heartbeat resolution.

Advanced UI/HUD
Developing a tablet-native HUD for field responders, providing real-time 3D occupancy heatmaps instead of raw Doppler stats.



References

- Why do People Die in Earthquakes: https://openknowledge.worldbank.org/bitstreams/d667ab90-a0f1-575d-83d6-41d4dafce08d/download
- Golden 72 Hour rule: https://www.cs.nccu.edu.tw/~lien/Pub/c92ccn.pdf
- Fire deaths in the world: https://www.fire-smi.ru/jour/article/view/772/0?locale=en_US
- Firefighter fatalities in the US: https://apps.usfa.fema.gov/firefighter-fatalities/
- Xavier 400: https://camero-tech.com/xaver-products/xaver-400/
- L-3 CyTerra: https://vcasecurity.com/product/l-3-cyterra-range-r/
- ADALM-Pluto: https://www.analog.com/media/en/news-marketing-collateral/product-highlight/adalm-pluto-product-highlight.pdf

References

- https://www.originwirelessai.com/
  Smart devices transformed into virtual sensors
- https://www.sify.com/technology/this-neural-network-uses-wifi-to-see-through-walls/
  Nueral network using WiFi to sense through walls
- https://www.universityofcalifornia.edu/news/how-wi-fi-can-give-us-x-ray-vision
  Theoretical and practical explanation of CSI Sensing
- https://spinoff.nasa.gov/FINDER-Finds-Its-Way-into-Rescuers-Toolkits
  NASA FINDER, human detection through concrete
- https://www.youtube.com/watch?v=M1E3DLwGHF4
  Further NASA FINDER data.
- https://youtu.be/WiTQxNaKAYA?si=ZAyVYqNiomqk9z0E
  Doppler Effect Calculations

References

- https://www.everythingrf.com/community/what-is-a-fmcw-radar FCMW explanation
- https://ieeexplore.ieee.org/document/10599163 Detecting Human Presence using FMCW wave radar
- https://pmc.ncbi.nlm.nih.gov/articles/PMC3008652/ Golden Hour catastrophe data

References (Datasheets)

- https://www.amazon.com/Dual-Band-Directional-Antenna-Wireless-Distance/dp/B0BWM88FRB?th=1= 12 DBi 2.4GHz-5.8GHz bi-directional antennas
- https://www.minicircuits.com/pdfs/VBFZ-5500-S+.pdf VBFZ-5500-S+ Band Pass Filter
- https://wiki.analog.com/resources/eval/user-guides/circuits-from-the-lab/cn0523 505-EVAL-CN0523-EBZ-ND Amplifier
- https://wiki.analog.com/resources/eval/user-guides/circuits-from-the-lab/cn0534 505-EVAL-CN0534-EBZ-ND Low Noise Amplifier